International Journal of Forecasting 41 (2025) 631–648

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

Humans vs. large language models: Judgmental forecasting in an era of advanced AI

Mahdi Abolghasemi [a,*], Odkhishig Ganbold [b], Kristian Rotaru [c,d]

[a] School of Mathematical Sciences, Queensland University of Technology, Brisbane, QLD, Australia
[b] Department of Medicine at the Royal Melbourne Hospital, Melbourne Medical School, University of Melbourne, Australia
[c] Department of Accounting, Monash Business School, Monash University, Australia
[d] BrainPark, Turner Institute for Brain and Mental Health, School of Psychological Sciences and Monash Biomedical Imaging Facility, Monash University, Australia

ARTICLE INFO

Article history:
Dataset link: https://github.com/mahdiabolghasemi/LLM-vs.-Humans-Forecasting

Keywords:
Judgmental forecasting
Large language models
Promotion
Sales forecasting
Forecast accuracy
Controlled experiment

ABSTRACT

This study investigates the forecasting accuracy of human experts versus large language models (LLMs) in the retail sector, particularly during standard and promotional sales periods. Utilizing a controlled experimental setup with 123 human forecasters and five LLMs—namely, ChatGPT-4, ChatGPT3.5, Bard, Bing, and Llama2—we evaluated forecasting precision through the absolute percentage error. Our analysis centered on the effect of the following factors on forecasters' performance: the supporting statistical model (baseline and advanced), whether the product was on promotion, and the nature of external impact. The findings indicate that LLMs do not consistently outperform humans in forecasting accuracy and that advanced statistical forecasting models do not uniformly enhance the performance of either human forecasters or LLMs. Both human and LLM forecasters exhibited increased forecasting errors, particularly during promotional periods. Our findings call for careful consideration when integrating LLMs into practical forecasting processes.

© 2024 The Authors. Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

Retail supply chain optimization and inventory management critically depend on the precision of demand forecasting. Effective forecasting integrates two principal components: quantitative models and expert human judgment (Brau, Aloysius, & Siemsen, 2023; Perera, Hurley, Fahimnia, & Reisi, 2019). Quantitative forecasting has achieved a level of sophistication and is systematically implemented in various forecasting solutions; however, the nuance of human judgment remains less quantifiable and is not as thoroughly researched (Sroginis, Fildes, & Kourentzes, 2023). Human expertise plays a pivotal role at several points in the forecasting continuum—ranging from data selection to model optimization, and most notably, in the application of discretionary adjustments to model outputs (Arvan, Fahimnia, Reisi, & Siemsen, 2019; Lawrence, Goodwin, O'Connor, & Önkal, 2006; Perera et al., 2019). Such expert-informed adjustments are suggested to improve forecast accuracy by integrating qualitative market insights with quantitative predictions—insights often elusive to purely algorithmic approaches (Fildes, Goodwin, Lawrence, & Nikolopoulos, 2009; Lawrence et al., 2006). However, there have been limited efforts to systematically explore in controlled experimental settings the mechanisms and reasoning behind experts' modifications of forecasts, and specifically how the integration of advanced forecasting methods and access to qualitative information about past and upcoming promotional periods may influence the precision of

* Corresponding author.
E-mail addresses: mahdi.abolghasemi@qut.edu.au (M. Abolghasemi), odkhishig.ganbold@unimelb.edu.au (O. Ganbold), kristian.rotaru@monash.edu (K. Rotaru).

https://doi.org/10.1016/j.ijforecast.2024.07.003
0169-2070/© 2024 The Authors. Published by Elsevier B.V. on behalf of International Institute of Forecasters. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

expert-informed adjustments (Fildes, Goodwin, & Önkal, 2019; Sroginis et al., 2023).

The integration of generative AI and large language models (LLMs) into forecasting represents a notable development, directly following the above discussion on the role of human expertise in enhancing statistical forecasts. LLMs are complex computational models trained on large-scale datasets, capable of generating remarkably human-like text (Brown et al., 2020; Chang et al., 2023). As LLMs like GPT can analyze vast amounts of qualitative information well beyond the capabilities of human forecasters, integrating LLMs into forecasting processes will likely enhance performance in the near future (Kirshner, 2024). Their ability to process and analyze large volumes of data and provide contextually relevant information has sparked interest in their potential to enhance judgmental forecasting (Makridakis, Petropoulos, & Kang, 2023; Schoenegger, Park, Karger, & Tetlock, 2024). Such AI models are equipped to absorb and recall information, propose actionable insights, and engage in reasoning processes that may rival human cognitive abilities (Xue, Voutharoja, & Salim, 2022; Yu, Chen, Ling, Dong, Liu, & Lu, 2023). Despite their potential, the performance of LLMs in judgmental forecasting, particularly during promotional periods characterized by complex endogenous and exogenous variables, is not well established (Makridakis et al., 2023). Consequently, the comparison of forecasting performance between LLMs and human forecasters emerges as an important research question (Halawi, Zhang, Yueh-Han, & Steinhardt, 2024; Kirshner, 2024; Schoenegger et al., 2024; Schoenegger, Tuminauskaite, Park, & Tetlock, 2024).

Recent studies in the field of forecasting indicate that LLMs could provide valuable predictive insights for economic and market forecasting (Kim, Muhn, & Nikolaev, 2024; Yu et al., 2023; Zhao et al., 2024). These models have shown a level of understanding and capability that is noteworthy in these complex domains. A study by Li, Feng, Yang, and Huang (2024) specifically explores the potential of ChatGPT as a financial advisor, examining its ability to forecast listed firm performance. Their findings suggest that ChatGPT can correct the optimistic biases of human analysts, demonstrating the potential of LLMs to offer more balanced and accurate forecasts in financial contexts. This aligns with the broader trend of employing advanced analytical methods with LLMs, potentially enhancing the forecasting process by leveraging their extensive knowledge base and inferential reasoning. However, recent empirical evidence also highlights certain limitations of LLMs compared to human forecasters, particularly in tasks that require probabilistic forecasting of future events. Notably, Schoenegger and Park (2023) found that in a real-world forecasting tournament, GPT-4's probabilistic forecasts were significantly less accurate than those made by human participants. This underperformance of LLMs in scenarios requiring advanced decision-making and adaptive reasoning suggests that while LLMs like ChatGPT can offer advantages in specific areas, such as mitigating human biases in financial forecasting, their capabilities in more dynamic and uncertain forecasting scenarios remain limited.

This study aimed to methodically assess the inherent capabilities of LLMs in judgmental forecasting and to compare their performance with that of human experts. In controlled experiments, both LLMs and human forecasters operated without any prior training or feedback on forecast accuracy, following the methodology used in studies such as Schoenegger et al. (2024). Consequently, our research directly evaluates the default capabilities of LLMs, providing an unadulterated assessment of their judgment and decision-making processes under experimental conditions. We explore the performance of LLMs across various scenarios, including their use alongside both simple baseline and advanced statistical models, their effectiveness during sales promotions, and their responses to external variables that add complexity to forecasting tasks.

By calculating and comparing the absolute percentage error (APE) of forecasts produced by human experts and a selection of LLMs, including the advanced ChatGPT4, we aim to contribute new insights into the efficacy of AI-driven models in enhancing forecasting accuracy. This method aligns with the approach advocated by Sroginis et al. (2023), who support the use of controlled, laboratory-based experiments as a means to obtain an accurate assessment of judgmental adjustments in the forecasting process. Indeed, laboratory experiments have been instrumental in demonstrating the importance of contextual understanding and product-specific knowledge, which extend the accuracy of forecasts beyond the capabilities of statistical models alone (Arvan et al., 2019; Sroginis et al., 2023; Webby, O'Connor, & Edmundson, 2005). By doing so, we aim to provide a balanced view of the forecasting capabilities across both human and machine-based forecasters, offering a better understanding of the strengths and limitations inherent to each approach considering the current level of development of LLMs.

The findings of our study present a comparison between human forecasters and LLMs in terms of their forecasting accuracy measured by APE. Our analysis indicates that LLMs do not uniformly exceed the predictive accuracy of human forecasters. Specifically, the most recent models, such as ChatGPT4 and Bing, showed a level of performance comparable to that of humans, whereas earlier models, including ChatGPT3.5, Bard, and Llama2, registered higher APEs. This suggests variability in the ability of different LLMs to process large datasets and produce accurate forecasts, possibly due to unique model limitations or the need for more sophisticated training to cope with diverse sales environments. A closer examination of the data indicates varying degrees of response to sales data, forecast type, promotional periods, and external factors, by both human forecasters and LLMs.

This research enriches the forecasting literature by providing an analysis that compares the performance of human forecasters with that of LLMs across diverse conditions, emphasizing the sophistication of forecasting models, the sales context, and external influences as critical factors in performance assessment. The study examines the capacity of LLMs to interpret information amidst promotions and external variables, offering a comprehensive

evaluation of these models' roles in complex forecasting situations. The outcomes of this research have substantial implications for the enhancement of demand planning and the design of advanced forecasting support systems that more seamlessly incorporate human expertise. Furthermore, our work opens new avenues for research into the integration of LLMs with existing forecasting practices and the refinement of collaborative human–AI decision support frameworks.

The remainder of this paper is organized as follows: Section 2 reviews relevant literature and formulates the hypotheses; Section 3 outlines the methodology and describes the participants; Section 4 presents the findings; Section 5 discusses these results and concludes with a reflection on the study's limitations and directions for future research.

2. Background and hypothesis development

2.1. Background and literature review

In sales forecasting, the integration of statistical models, advanced machine learning algorithms, and human judgment has become increasingly sophisticated. The evolution of forecasting models from simple baseline to advanced integrated systems reflects this complexity. Baseline models, effective at capturing fundamental trends, often struggle in volatile scenarios like promotions (De Baets & Harvey, 2018). Advanced models, incorporating techniques such as dynamic linear regression, provide a robust solution by integrating complex factors like promotional impacts and pricing strategies (Abolghasemi, Tarr, & Bergmeir, 2022; Kourentzes & Petropoulos, 2016; Ma, Fildes, & Huang, 2016). This progression addresses the cognitive challenges forecasters face and highlights the need for balancing statistical rigor with human judgment (Fildes & Goodwin, 2007; Fildes et al., 2009).

Human judgment, essential in adjusting statistical forecasts, particularly in promotional contexts, introduces complexities and biases impacting forecast quality (Goodwin & Wright, 2010; Sroginis et al., 2023). In promotional scenarios, the complexity and unpredictability inherent to consumer behavior necessitate a judicious combination of model-based forecasts and expert intuition (Fildes & Hastings, 1994; Sroginis et al., 2023). The significance of human judgment in forecasting, particularly in scenarios influenced by promotions, represents a crucial field of inquiry in both academic research and practical application. The effectiveness of human intervention in these settings exhibits considerable variability, as evidenced by studies such as Kourentzes and Fildes (2023), Trapero, Pedregal, Fildes, and Kourentzes (2013), and Fildes and Goodwin (2007).

The cognitive load associated with complex forecasts can result in suboptimal decision-making, highlighting the need for more intuitive and supportive forecasting systems (Abolghasemi, Hurley, Eshragh, & Fahimnia, 2020; Lawrence et al., 2006; Lim & O'Connor, 1995). The literature shows that cognitive biases and heuristics significantly influence forecasters' judgments, often leading to systematic errors (Bolger & Harvey, 1993; Fildes, Ma, & Kolassa, 2022; Goodwin & Wright, 2010; Harvey, 1995). Thus, the trajectory of current research is steering towards a more integrated approach, where the strengths of human cognition are leveraged to complement and refine the outputs generated by advanced forecasting models (Kourentzes & Fildes, 2023; Sroginis et al., 2023). However, further exploration into the balance between human judgment and statistical models is needed to better understand the value added by human intervention.

The evolution from purely statistical to judgmental forecasting marks a shift towards integrating historical data with the context of current scenarios, aligning with the core objective of forecasting support systems. These systems aim to harmonize algorithm precision with human intuition to enhance overall forecast accuracy (Fildes & Goodwin, 2007; Green & Armstrong, 2015; Zhang, Patuwo, & Hu, 1998). However, there remains a gap in understanding the rationale behind judgmental adjustments, particularly the impact of special events, biases, and human factors on judgmental forecasting (Bolger & Önkal-Atay, 2004; Fildes & Goodwin, 2007; Lawrence et al., 2006; Sroginis et al., 2023). This gap calls for a deeper investigation into the cognitive processes involved in judgmental forecasting. Fildes et al. (2009) emphasized the complexity of these processes, revealing how biases and heuristics can influence forecasters' judgments. Bolger and Harvey (1993) explored how cognitive biases can lead to systematic errors, highlighting the need for systems that support and improve human judgment. Harvey (1995) emphasized the importance of understanding how individuals process and interpret information in forecasting. Goodwin and Wright (2010) showed how cognitive biases can influence judgmental forecasting, especially in complex or ambiguous situations. In essence, the progression towards integrating statistical and judgmental forecasting demands advanced analytical tools and a comprehensive understanding of the human element in forecasting.

The interaction between human cognition and statistical data needs further exploration to enhance forecast efficacy and accuracy, particularly in complex and uncertain environments. Our investigation explores both simple baseline forecasts and advanced promotional forecasts. Simple baseline forecasts effectively capture underlying time series trends and seasonality, yet frequently fail to account for volatile elements like promotions (De Baets & Harvey, 2018). In contrast, advanced baseline forecasts utilize dynamic linear regression models. These models are designed to address the need for a more robust forecasting tool that can assimilate the effects of promotional impacts and integrate key variables such as pricing strategies (Abolghasemi et al., 2020; Kourentzes & Petropoulos, 2016). The shift towards advanced forecasting models is a response to forecasters' limited cognitive resources. These models aim to reduce cognitive load and provide a more strategic overview of data for effective decision-making (Bolger & Harvey, 1993; Fildes, 1991; Fildes et al., 2009). The role of advanced models in mitigating the biases inherent in human judgment demonstrates their potential to enhance the accuracy and reliability of forecasts. The clarity provided by these models can be crucial in

avoiding the pitfalls of heuristic-driven decisions (Harvey & Bolger, 1996), which are more likely under conditions of excessive cognitive load. Thus, our exploration into the dichotomy of simple baseline versus advanced promotional forecasting models extends beyond a mere a comparative performance assessment; it is also an investigation into their potential to enhance human analytical capabilities and judgment in forecasting scenarios characterized by complexities, such as promotional periods.

The advent of LLMs like ChatGPT adds a new dimension to this research gap. Trained on extensive datasets and textual information, these models generate detailed, contextually relevant information, with potential applications in economic and market forecasting still under investigation (Makridakis et al., 2023; Schoenegger & Park, 2023). Their performance in complex scenarios, especially in promotional periods, is an emerging area of research. It is thus important to understand whether LLMs, as a new class of models, substantially change our knowledge of the performance of algorithmic models in judgmental forecasting or, put differently, whether LLMs outperform human forecasters, given a variety of critical factors that have been considered in judgmental forecasting literature. Below, we develop our theoretical predictions to contribute further to this investigation.

2.2. Hypothesis development

The study's hypotheses are directed towards evaluating the performance of LLMs relative to human judgment under various conditions, including simple baseline vs. advanced forecasts, promotional impacts, and external impact variables. This approach aligns with research emphasizing the complex dynamics between algorithmic models and human cognitive processes (Goodwin & Wright, 2010; Lawrence et al., 2006). We also propose hypotheses concerning the forecasting accuracy of humans and five distinct LLMs (assessing each forecaster individually) under these diverse circumstances. To address the potential confounding effects of actual sales figures on our forecast accuracy metric (i.e., the absolute percentage error), we incorporated actual sales as a control variable in our analysis. This adjustment is crucial for mitigating the potential confounding effects of fluctuations in sales figures, which can skew the performance metrics of forecasters. By controlling for actual sales, we aim to isolate the true impact of the variables under investigation, providing a more precise estimation of how various conditions directly affect forecasting accuracy. This ensures that our results more accurately reflect the specific effects of the variables of interest, rather than being influenced by the actual sales volatility.

2.2.1. Comparative accuracy of LLMs and human forecasters (H1)

The adoption of LLMs in time series forecasting prompts a critical evaluation of their performance compared to human forecasters. Human forecasters integrate a broad spectrum of experiential and contextual knowledge into their predictions and often employ heuristics to simplify complex tasks. However, their judgments are vulnerable to several well-documented cognitive biases such as overconfidence, anchoring, and confirmation biases, which impair the accuracy of their forecasts (Chapman & Johnson, 1999; Lawrence et al., 2006; Tversky & Kahneman, 1974). LLMs, on the other hand, leverage vast amounts of data and sophisticated pattern recognition algorithms to generate forecasts, which may enhance consistency and reduce susceptibility to these biases. Nevertheless, LLMs can also manifest new biases, such as hallucinations, which adversely impact their performance (Brown et al., 2020; Silver et al., 2016).

In the domain of sales forecasting, achieving accuracy is crucial, and the ongoing debate pits human expertise against algorithmic precision. Human forecasters bring important insights into market dynamics and can swiftly adapt to new information, but their effectiveness is often hampered by psychological biases and a limited capacity to process large data volumes (Lawrence et al., 2006). In contrast, LLMs, unencumbered by these psychological limitations and adept at processing extensive datasets, present a promising alternative. With their ability to discern trends and patterns from comprehensive data (Silver et al., 2016), LLMs are hypothesized to yield more accurate judgmental forecasts than human forecasters, especially in complex forecasting scenarios such as promotional sales. This hypothesis anticipates a discernible difference in forecasting performance, favoring LLMs over humans, in terms of accuracy.

Hypothesis 1. LLMs will surpass human forecasters in forecasting accuracy when adjusted for comparable levels of actual sales.

2.2.2. Enhanced forecasting performance with advanced forecast models (H2)

In exploring the impact of advanced promotional models on forecasting accuracy (e.g. Sroginis et al., 2023), it is important to consider how additional information in the form of expert predictions or algorithmic projections may influence the decision-making processes of both human forecasters and LLMs. Research in judgmental forecasting suggests that humans often benefit from having access to advanced forecast information, which can serve as an anchor and potentially improve accuracy by reducing the range of possible outcomes considered (Goodwin & Fildes, 2022; Lawrence et al., 2006). However, humans may also overly rely on this information and underweight their own judgment (Meub & Proeger, 2016).

For LLMs, advanced forecasting models can provide accurate data to be integrated into the LLMs' reasoning process, potentially enriching the model's inputs and leading to better performance, especially when these advanced forecasts encapsulate complex patterns that the simple baseline models alone might not detect. Nevertheless, the effectiveness of this integration depends on the quality of the forecasts and the ability of the LLMs to appropriately weigh this information against the data it has already processed.

As discussed in our literature review and in the lead to Hypothesis 1, the provision of advanced forecasts is likely to influence the accuracy of subsequent predictions made

by both human forecasters and LLMs. For human forecasters, research has highlighted how the use of advanced forecasts can serve as a heuristic, aiding in the process of making complex judgments and potentially leading to improved accuracy, though it also carries the risk of over-reliance and the suppression of independent judgment (Goodwin, 2005; Lawrence et al., 2006). On the other hand, for LLMs, which are designed to synthesize large volumes of data and identify patterns, advanced forecasts could provide an additional layer of information, enhancing the model's predictive capabilities (Makridakis et al., 2023). The advanced forecasts, if accurately capturing future trends, could therefore improve the LLM's predictive performance by offering a form of 'expert input' that the model can incorporate into its algorithms effectively. This integration is not influenced by the psychological biases that affect human forecasters, potentially leading to a more optimal use of the provided advanced forecasts.

Given the above considerations, we hypothesize the following:

Hypothesis 2. Both human forecasters and LLMs will exhibit improved forecasting performance when provided with advanced forecasts, due to the additional information serving as a data-rich input that can aid in refining predictions (H2a). However, the degree of improvement will differ between the two groups, with LLMs expected to utilize advanced forecasts more effectively, as they can systematically integrate and evaluate such inputs without the cognitive biases that human forecasters are subject to (H2b).

2.2.3. Variability in forecast accuracy during promotional and non-promotional periods (H3)

Building upon the challenges posed by promotional periods in sales forecasting, our theoretical prediction hinges on the systematic data processing capabilities of LLMs. The proficiency of LLMs at forecasting during promotional periods, however, is predicated on their exposure to a diverse set of historical promotional data. Such data enable these models to adjust to the volatility introduced by promotions more consistently than humans, who might underreact or overreact to the influence of promotions on consumer behavior. While the LLMs were not explicitly trained on promotional period data in this study, their inherent capabilities may already include the ability to anticipate the impact of promotional activities on sales. This potential understanding could result in more consistent forecasting accuracy across promotional and non-promotional periods compared to human forecasters, as we test our predictions.

Thus, Hypothesis 3 is formulated with the expectation that LLMs, given their algorithmic nature and assuming comprehensive training on promotional data, will demonstrate a less pronounced difference in forecasting accuracy between promotional and non-promotional periods compared to human forecasters, whose judgment may be clouded by inherent cognitive biases:

Hypothesis 3. The accuracy of both human forecasters and LLMs will differ during promotional versus non-promotional periods (H3a), with LLMs expected to demonstrate a less pronounced difference in accuracy between these periods compared to humans (H3b).

2.2.4. Impact of external conditions on forecasting disparities (H4)

Forecasting in the face of external impacts is a complex endeavor, requiring the consideration of diverse, often non-linear influences on sales data. Understanding the influence of external conditions on forecasting accuracy necessitates distinguishing between negative and positive impacts. In our study, external impact is presented as a compound effect of competitors' activity such as advertisements, indicating whether such activity has impacted or will impact product sales positively or negatively. These conditions introduce volatility into sales data, complicating the forecasting process (Abolghasemi, Beh, Tarr, & Gerlach, 2020).

Human forecasters, when integrating and interpreting external impact information—often qualitative in nature—can exhibit significant biases influenced by the emotional valence of the information, whether negative or positive (Sroginis et al., 2023). During economic downturns or natural disasters, for instance, forecasters might disproportionally emphasize such negative information, resulting in overly conservative estimates due to a well-documented negativity bias (Ito, Larsen, Smith, & Cacioppo, 1998; Lewandowsky, Ecker, Seifert, Schwarz, & Cook, 2012). Conversely, during periods of positive economic outlooks or favorable holiday seasons, forecasters may demonstrate undue optimism in their predictions (Eroglu & Croxton, 2010; Goodwin & Fildes, 2022; Önkal, Gönül, & Goodwin, 2023). Negative information often influences evaluations more strongly than comparably significant positive information and is a recognized phenomenon in expert forecasts (Chang & Hao, 2022). Such asymmetry in human judgment, often driven by affect and representativeness heuristics, leads to greater variability in forecasting accuracy, particularly during periods of negative versus positive external impact.

In contrast, LLMs, devoid of emotional responses, analyze data patterns algorithmically. Provided that their default training includes a range of both negative and positive external impacts, LLMs can maintain a consistent forecasting approach. They adjust to different conditions based on learned data patterns rather than emotional responses, a capability rooted in their algorithmic nature (Chang et al., 2023). Therefore, while both human forecasters and LLMs are likely to experience variances in forecasting accuracy due to external impacts, the extent of these variances may differ. Humans are expected to show higher sensitivity to the valence of external information due to emotional and cognitive biases. LLMs, unaffected by such biases, are anticipated to exhibit a smaller disparity in APE across varying external conditions.

Hypothesis 4. The accuracy of both human forecasters and LLMs will differ under negative versus positive external impact conditions (H4a). Human forecasters will show a greater difference in accuracy under negative versus positive external impact conditions (i.e., lower accuracy) than LLMs (H4b).

Table 1
Treatment conditions tested in this study.

Treatment condition	Promotion	Statistical model	External impact
1	Yes	Baseline	Positive
2	Yes	Advanced (promotional)	Positive
3	Yes	Baseline	Negative
4	Yes	Advanced (promotional)	Negative
5	No	Baseline	Positive
6	No	Baseline	Negative

2.2.5. Variability in forecast accuracy under complex forecasting scenarios (H5)

The dynamic environment of sales forecasting, where numerous factors intersect, may impose challenges for human forecasters to maintain high levels of accuracy due to the cognitive strain imposed by the need to balance and interpret multiple, and sometimes conflicting, pieces of information. The propensity for cognitive biases and the potential for information overload may lead to increased variability in accuracy among human forecasters when considering all factors simultaneously.

When considering a comprehensive forecasting model that accounts for all previously considered factors—such as the presence of advanced forecasts, promotional versus non-promotional periods, and negative versus positive external impacts (forming therefore six treatment conditions, see Table 1)—the interaction between these variables becomes crucial in determining forecasting accuracy. This multifaceted approach reflects the real-world complexity of sales forecasting, where multiple factors often simultaneously influence outcomes.

LLMs are designed to handle large volumes of complex data, potentially allowing them to better navigate the interactions between multiple forecasting factors without succumbing to the biases that affect human judgment (Chang et al., 2023). Their algorithmic efficiency and immunity to psychological biases allow them to systematically analyze complex interactions between factors. Therefore, when all factors from the forecasting model are taken into account, LLMs are hypothesized to achieve more consistent and accurate performance compared to human forecasters, as indicated by a generally higher forecasting accuracy that is less affected by changes in conditions.

Hypothesis 5. The accuracy of human forecasters will vary across treatment conditions that incorporate advanced forecasts, promotional conditions, and external impacts (H5a). When considering a model that incorporates the aforementioned treatment conditions, LLMs will demonstrate more accurate forecasting across these conditions, as indicated by a consistently higher accuracy, compared to humans (H5b).

3. Method

3.1. Participants

The experimental study was conducted in August 2023 with 123 human forecasters and five LLMs. The human forecasters were business school graduate students (65.9% female, 34.1% male) who voluntarily participated in this study conducted in the Monash Business Behavioural Laboratory, Monash University, Australia. The average age of the human forecasters was 24.21 years (SD = 2.51 years), ranging from 22 to 34 years old, and they had on average 1.46 years (SD = 1.60 years) of business experience. All human forecasters had practical skills in business forecasting, since they all completed a unit on time series forecasting where they had to demonstrate practical skills in applying a variety of time series forecasting models for stationary and non-stationary data (including data with additive or multiplicative seasonal effects).

In addition, we used five popular and powerful LLMs: ChatGPT3.5 and ChatGPT4 developed by OpenAI, Llama2 developed by Meta, Bing developed by Microsoft, and Bard developed by Google. These models were among the most advanced natural language processing models at the time when the experiment was conducted. They benefit from at least millions of parameters, they are trained on a vast amount of data, and fine-tuned to reason, answer questions, and generate text in a human-like manner.

3.2. Design

The study's design was a 2 × 2 + 2 mixed design, involving three independent variables (IVs). The first IV, 'Statistical forecast', had two conditions: a 'Baseline' and an 'Advanced' model. The second IV, 'External impact', differentiated between positive and negative effects. The third IV, 'Promotion', was divided into 'Yes promotion' and 'No promotion' conditions. Notably, the advanced statistical forecast, which accounted for promotional effects, was exclusively applied when promotions were active. Consequently, this design yielded six experimental conditions (as per Table 1):

1. Yes promotion, Positive external impact, Baseline statistical forecast.
2. Yes promotion, Positive external impact, Advanced statistical forecast.
3. Yes promotion, Negative external impact, Baseline statistical forecast.
4. Yes promotion, Negative external impact, Advanced statistical forecast.
5. No promotion, Positive external impact, Baseline statistical forecast.
6. No promotion, Negative external impact, Baseline statistical forecast.

Each participant evaluated the same collection of 24 time series. The accuracy of the participants' forecasts was

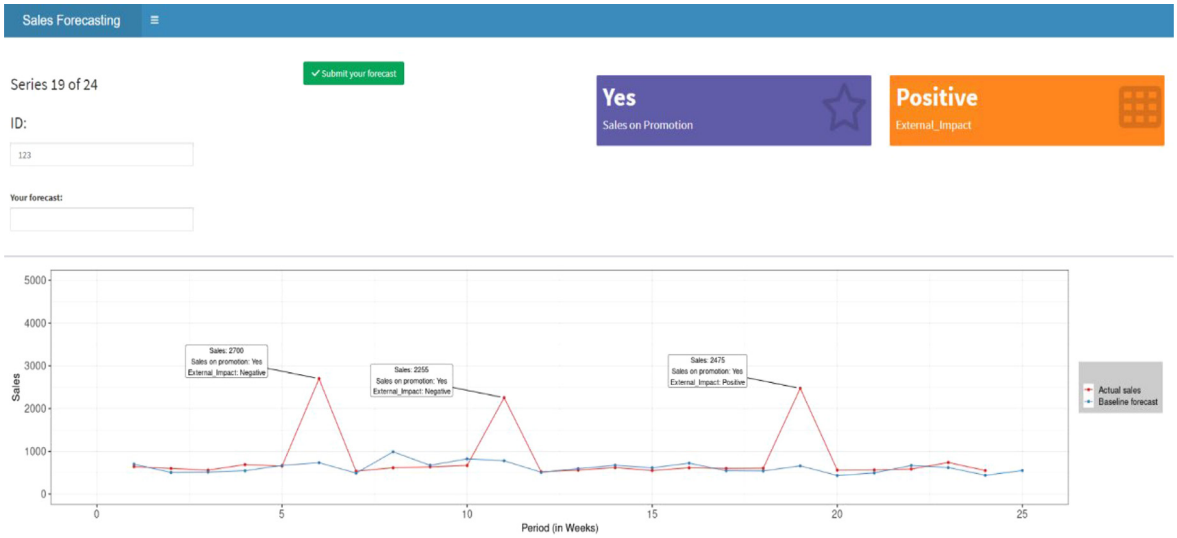


Fig. 1. Interface of the experiment dashboard.

gauged using the absolute percentage error (APE) as the dependent variable (DV). The APE quantifies forecasting precision by comparing the forecasted values against actual sales, presented as a percentage. The lower the APE value, the more accurate the forecast. This measure is critical for assessing the effectiveness of judgmental forecasting against statistical models under varying market conditions and promotional strategies.

Based on their performance (forecasting accuracy), the participants were awarded between $5 and $20 AUD. The forecasting accuracy was calculated based on the APE as follows:

$$APE = \frac{abs\,(f_t - x_t)}{x_t} \quad (1)$$

where f_t is the forecast at time t, and x_t is the actual sales at time t.

3.3. Materials and apparatus

The experiment was designed with R's 'Shiny' package and hosted on RStudio.[1] Participants were given 24 different sales series representing the actual sales of different products in an Australian company. The company is a food manufacturing company that produces health and breakfast products and distributes their products through two main retailers in Australia. Each series included 24 observations, varying in sales volume, promotion status, and the extent of sales uplifts resulting from promotions.

Fig. 1 shows a screenshot of the experiment graphical interface. It represents the dashboard of the forecasting system that participants used to conduct their forecasting decisions.

'Promotion'—the purple box at the top-right of the dashboard—is the main variable that causes sales uplifts. It is a binary variable that takes either 'Yes promotion' or 'No promotion', representing the presence and absence of a promotion, respectively. The orange box represents the 'External impact' variable, a binary indicator summarizing the effects of competitors' activities on sales. It takes either 'Positive' or 'Negative' values to signify an increasing or decreasing impact on sales volume, respectively. This impact may vary between promotional and non-promotional periods and influences demand by marginally increasing or decreasing sales volumes up to 10%. This approach was influenced by a case study that examined how a company accounts for promotional impacts on sales while also considering competitor activities. The company's forecasts leverage continuous market intelligence, enabling specialists to adjust their predictions in response to comprehensive insights about competitors' policies, prices, and promotions. The influence of the 'External impact' variable on sales is dual: it is perceived negatively if competitors promote similar products when the company's products are not on major sale, and positively if the company's products are on major sale while competitors' products are not. While these impacts are specific to individual products, they are generalized by the company's experts based on overall market share and conditions.

Besides this information, participants were given a statistical forecast, depicted in the blue dotted line on the main dashboard. The statistical forecast may be one of two types: a simple baseline statistical forecast or an advanced statistical forecast. The simple baseline forecast is a forecast generated by the forecast support system of the company, which uses Holt–Winter exponential smoothing model to forecast baseline sales in the absence of promotions and does not consider promotions or the external impact. The advanced forecast takes into account the impact of promotions but not the external impact, and it uses a dynamic linear regression model to forecast sales.

The given information includes two types of dotted lines, i.e. blue representing historical forecasts and red denoting historical sales for the last 24 periods of the

[1] The task can be accessed here: https://mahdiforecasting.shinyapps.io/exp1/

time series, depicted on the dashboard. We provided the same graph and dashboard for all participants to avoid any interfering impact of graph types (Reimers & Harvey, 2023). Other relevant information provided to the participants included the promotion status and external impact in Period 25, and the statistical forecast (simple baseline or advanced baseline statistical forecast). We assume that the given information sufficiently encapsulated all internal and external factors influencing sales.

3.4. Procedures

Participants (human forecasters and LLMs) were instructed to assume the role of the sales manager in a hypothetical company. In this role, they were tasked with forecasting sales for Period 25 based on the provided information. Each participant evaluated the same collection of 24 separate time series. These series represented genuine sales data for various health and breakfast products from an Australian food manufacturer, encompassing fluctuations in sales volume, promotional activities, and sales boosts from promotions. Period 25 is either a promotion period or non-promotion period. For simplicity, participants were not asked to forecast any immediate post-promotion periods.

The ability of LLMs to generate accurate responses heavily depends on the given prompts. Prompts are texts that human can give to LLMs to provide them with the context and ask them for responses. To ensure fairness in our experiment, we provided the LLMs with the same information that was given to the human participants. This was done during the same week in August 2023 when the experiment with the human forecasters was conducted. In our prompt engineering, the data provided effectively characterized the task for the LLMs. This information comprised handouts, time series observations in tabular format, and additional details such as promotion status, statistical forecasts, and external impact information, all of which were provided to the human participants. The precise prompt provided to the LLMs is given in Appendix B. The same prompt was used for all LLMs considered in this study. Detailed descriptive statistics for the experimental data are provided in Appendix C.

4. Results

We employed both mixed-effects models and regression models to test our hypotheses, with the latter specifically used to evaluate the performance of each LLM individually. Following the approach of Fildes et al. (2019), we utilized linear mixed-effects models with restricted maximum likelihood for our statistical analysis. This method is particularly advantageous for handling correlated repeated-measures data and time-varying covariates due to its flexibility, robustness against missing data, and ability to model realistic variance and correlation patterns (Cnaan, Laird, & Slasor, 1997; Gueorguieva & Krystal, 2004).

The general equation for our mixed-effects models is as follows:

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + \beta_2 Z_{ij} + u_j + \epsilon_{ij},$$

where

- Y_{ij} represents the dependent variable (APE) for the i-th observation within the jth forecaster,
- β_0 is the intercept,
- X_{ij} is the covariate for actual sales, reflecting its impact on APE,
- β_1 is the coefficient for actual sales,
- Z_{ij} encompasses the categorical fixed effects from various experimental conditions, including forecaster type (humans and LLMs), forecast model (baseline or advanced), promotional status (promotion or non-promotion), and external impact (positive or negative), along with their interactions,
- β_2 denotes coefficients for the fixed effects,
- u_j is the random effect for each forecaster, capturing unexplained variability across forecasters, and
- ϵ_{ij} is the residual error for each observation.

In our analysis, actual sales are included as a covariate to mitigate the influence of sales volume variations on the APE. Experimental conditions—such as promotion status ('Yes promotion', 'No promotion'), forecast type ('Baseline forecast', 'Advanced forecast'), external impact ('Positive external impact', 'Negative external impact'), and treatment interventions (the interaction of the aforementioned conditions)—are integrated as fixed effects. This structure helps clarify how different forecasting contexts affect forecast accuracy. By treating each forecaster as a random effect, our model adapts to individual differences among forecasters, thus enhancing the analysis depth. This approach not only captures the variability inherent in real-world forecasting but also boosts the generalizability and reliability of our findings.

When testing hypotheses for each forecaster type, where each condition is represented by a single sample observation (i.e., each LLM forecaster), we employ regression models because the data structure does not support testing for random effects, which are a requisite for the application of mixed-effects models. We discuss the main results in the following sections. Descriptive statistics of forecast accuracy for both human and LLM forecasters are provided in Appendices D and E. Additionally, more comprehensive results on H2a–H5a are detailed in Appendix J.

4.1. Comparative accuracy of LLMs and human forecasters (H1)

Hypothesis 1 posited that LLMs would surpass human forecasters in terms of forecasting accuracy, as measured by APE, under the assumption of comparable levels of actual sales. The premise was that LLMs, by virtue of their capacity to process extensive datasets and discern complex patterns devoid of human cognitive biases, would provide a more data-driven and objective forecasting approach.

Contrary to this hypothesis, the results of the mixed-effects regression analysis revealed a more nuanced picture (see Table 2 and Fig. 2). The performance of the LLMs varied, with ChatGPT3.5 and Llama2 yielding significantly higher APE values compared to human forecasters, indicating inferior forecasting accuracy.

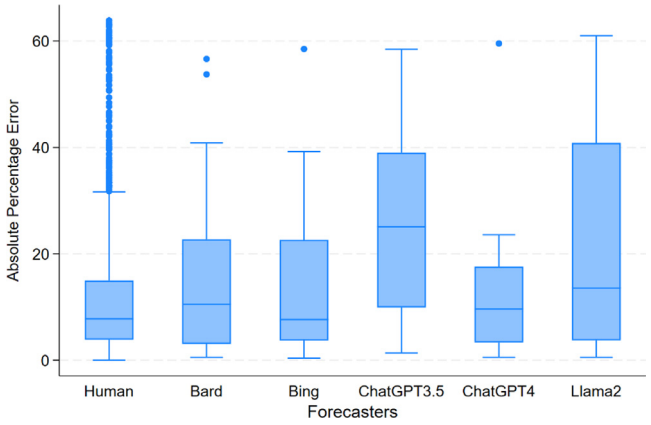


Fig. 2. APE by forecaster type: Humans vs. LLMs.

Table 2
Mixed-effect models results for H1.

	APE		
	Coefficient	Std. err.	P-value
Fixed effects			
Intercept	14.35	0.54	<.001
Actual sales	−0.001	0.002	<.001
Bard	4.89	2.85	.086
Bing	2.73	2.69	.311
ChatGPT3.5	14.77	2.80	<.001
ChatGPT4	0.69	2.64	.794
Llama2	9.18	2.92	.002
Random effects			
	Estimate	Std. err.	
Forecaster	1.15	0.94	
Residual	138.77	3.68	
Observations	2973		

On the other hand, Bard, Bing, and ChatGPT4 did not show significant differences in APE compared to human forecasters, which indicates that their forecasting accuracy was not notably worse or better than that of humans. These findings challenge the initial hypothesis, revealing that not all LLMs are inherently better at forecasting than humans and that the efficacy of LLMs can be model-specific. Therefore, while some LLMs may have the theoretical capability to outperform human forecasters, the empirical evidence suggests that this capability may not always translate into better forecasting accuracy in practice.

4.2. Enhanced forecasting performance with advanced forecast models (H2)

Considering H2a, which posits that both human forecasters and LLMs would benefit from advanced forecasts, separate models were tested for each forecaster type. In doing so, we conducted a mixed-effects regression analysis for human forecasters to account for within-subject variability and separate regression analyses for each LLM. The results are shown in Appendix J. In the mixed-effects regression analysis, with baseline versus advanced forecasting methods as predictors, advanced forecasts did not significantly reduce accuracy for human forecasters compared to baseline forecasts. Similarly, advanced forecasts did not yield any significant changes in forecast accuracy for each of the LLMs when compared to baseline forecasts. In H2b, we hypothesized that LLMs will show a greater improvement in forecasting performance than human forecasters when using advanced forecasts. The results are shown in Table 3 and Fig. 3. Among the different types of forecasters, Bard, ChatGPT3.5, and Llama2 exhibited a significant increase in APE by 10.73%, 10.25%, and 9.79%, respectively, compared to the performance of human forecasters. This result does not support H2b, as these increases suggest a decline in forecasting performance when these LLMs are provided with advanced forecasts.

However, for Bard, the interaction term with the advanced forecast type significantly decreased APE by 14.64%, partially supporting H2b. This indicates that Bard performed better with advanced forecasts compared to its baseline performance. This interaction effect is noteworthy as it contrasts with the main effect of the advanced forecast type, suggesting that while advanced forecasts did not generally improve performance for forecasters, Bard was able to utilize the advanced information effectively. On the contrary, for ChatGPT3.5 interaction term with the advanced forecast type significantly increased APE by 11.85%, suggesting that ChatGPT3.5 performed worse when provided with advanced forecasts compared to its baseline performance.

4.3. Variability in forecast accuracy during promotional and non-promotional periods (H3)

To investigate the variability in APE across promotional and non-promotional periods for different types of forecasters (H3a), mixed-effects regression analysis for human forecasters and separate regression analyses for each LLM were conducted. The results are shown in Appendix J. The analysis results demonstrate that during promotional periods, human forecasters experienced a significant increase in APE, indicating reduced forecast accuracy. For the LLMs, ChatGPT3.5, and Llama2 showed a notable increase in APE during promotional periods, indicating

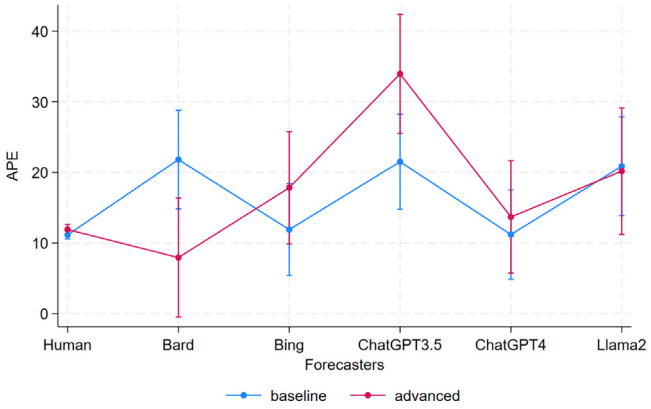


Fig. 3. Interaction: Forecasters and statistical forecast type.Note: The results are predictive margins with 95% CIs.

Table 3
Mixed-effects model results for H2.

	APE		
	Coefficient	Std. err.	P-value
Fixed effects			
Intercept	14.08	0.57	<.001
Actual sales	−0.001	0.0002	.001
Advanced forecast	0.81	0.46	.076
Bard	10.73	3.57	.003
Bing	0.67	3.33	.842
ChatGPT3.5	10.25	3.45	.003
ChatGPT4	0.05	3.23	.988
Llama2	9.79	3.57	.006
Advanced # Bard	−14.64	5.39	.007
Advanced # Bing	5.23	5.04	.300
Advanced # ChatGPT3.5	11.845	5.30	.025
Advanced # ChatGPT4	1.70	4.98	.732
Advanced # Llama2	−1.65	5.61	.769
Random effects			
	Estimate	Std. err.	
Forecaster	1.16	0.93	
Residual	138.24	3.67	
Observations	2973		

Table 4
Mixed-effects model results for H3.

	APE		
	Coefficient	Std. err.	P-value
Fixed effects			
Intercept	12.77	0.59	<.001
Actual sales	−0.001	0.0002	<.001
Yes promo	3.30	0.48	<.001
Bard	−0.70	4.56	.879
Bing	0.20	4.56	.965
ChatGPT3.5	−3.10	4.56	.496
ChatGPT4	0.56	4.56	.902
Llama2	−2.06	4.56	.651
Yes promo # Bard	8.87	5.45	.104
Yes promo # Bing	3.63	5.27	.491
Yes promo # ChatGPT3.5	26.95	5.38	<0.001
Yes promo # ChatGPT4	0.14	5.23	.979
Yes promo # Llama2	18.15	5.53	.001
Random effects			
	Estimate	Std. err.	
Forecaster	1.42	0.94	
Residual	134.25	3.57	
Observations	2973		

a decline in forecasting performance compared to non-promotional periods. Conversely, Bard, Bing, and ChatGPT4 did not exhibit significant changes in APE during promotional periods.

Regarding H3b, the mixed-effects regression analysis indicated a significant overall increase in APE during promotional periods, implying reduced forecasting accuracy during these times. The results are shown in Table 4 and Fig. 4. Contrary to H3b, which predicted that LLMs would show a smaller difference in APE compared to humans, certain LLMs (ChatGPT3.5 and Llama2) experienced a significant increase in APE during promotional periods. This result suggests a decrease in their forecasting performance during promotional periods, contrary to the hypothesis.

The absence of significant main effects among different forecaster types, with coefficients ranging from −3.10 for ChatGPT3.5 to 0.56 for ChatGPT4, indicates that no single forecaster type consistently outperformed or underperformed human forecasters across conditions. Significant random effects in the model indicated considerable variability (1.42%) in APE among different forecasters. The inclusion of these random effects substantially improved the model's fit, as evidenced by the likelihood ratio test ($P < 0.05$).

4.4. Impact of external conditions on forecasting disparities (H4)

When addressing H4a, the mixed-effects analysis for human forecasters and regression analyses for each LLM were conducted in order to test the forecaster-specific APE under negative vs. positive external impact conditions. The results are shown in Appendix J. The parameters indicated substantial variability in forecasting performance among different forecasters. The results suggest that the presence of positive vs. negative external impacts on forecasting accuracy did not significantly vary by different forecaster types.

Our examination of the impact of negative versus positive external impacts on APE when comparing the forecast accuracy for LLMs versus human forecasters (H4b) yielded insights that partially counter H4b, as shown in Table 5

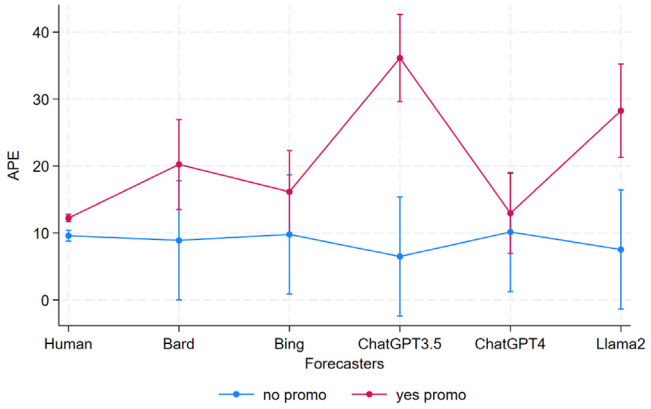


Fig. 4. Interaction: Forecasters and promotions.

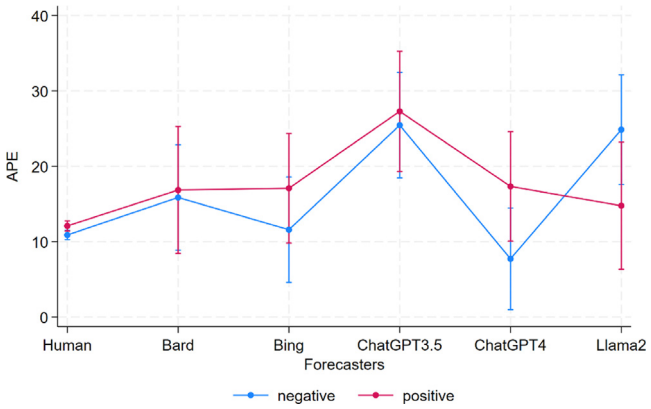


Fig. 5. Interaction: Forecasters and external impact.

and Fig. 5. Notably, the presence of a positive external impact was not found to be associated with a significant increase in APE. Contrary to the hypothesis, which anticipated a smaller difference in APE for LLMs, ChatGPT3.5 and Llama2 demonstrated significant increases in APE compared to human forecasters, while Bard, Bing, and ChatGPT4 did not exhibit significant changes. The interaction effects further suggested a complex response pattern, particularly for Llama2, with a decreased APE (i.e. improved accuracy) under positive impact. These results suggest that the algorithmic nature of LLMs did not uniformly mitigate the impact of positive external influences, as was hypothesized.

4.5. Variability in forecast accuracy under complex forecasting scenarios (H5)

In a series of regression analyses comprising mixed-effects model analysis for human forecasters and regression models for each LLM type, we examined the impact of various treatment conditions on APE for each forecaster type (H5a). In all models, the baseline condition, represented by the scenario 'Yes promotion, Positive impact, Baseline statistical forecast,' serves as the reference point for comparison. The results are summarized in Appendix J.

Table 5
Mixed-effects model results for H4.

	APE		
	Coefficient	Std. err.	P-value
Fixed effects			
Intercept	14.42	0.74	<.001
Actual sales	−0.001	0.0002	<0.001
Positive (env. impact)	−0.09	0.51	.859
Bard	4.93	3.58	.168
Bing	0.65	3.58	.855
ChatGPT3.5	14.52	3.58	<.001
ChatGPT4	−3.16	3.45	.360
Llama2	13.86	3.72	<.001
Positive # Bard	−0.11	5.39	.984
Positive # Bing	4.34	4.94	.379
Positive # ChatGPT3.5	0.57	5.21	.913
Positive # ChatGPT4	8.40	4.85	.083
Positive # Llama2	−11.12	5.49	.043
Random effects			
	Estimate	Std. err.	
Forecaster	1.15	0.94	
Residual	138.68	3.68	
Observations	2973		

For human forecasters, when compared to the baseline, the 'Yes promotion, Positive external impact, Advanced forecast' condition significantly increased APE.

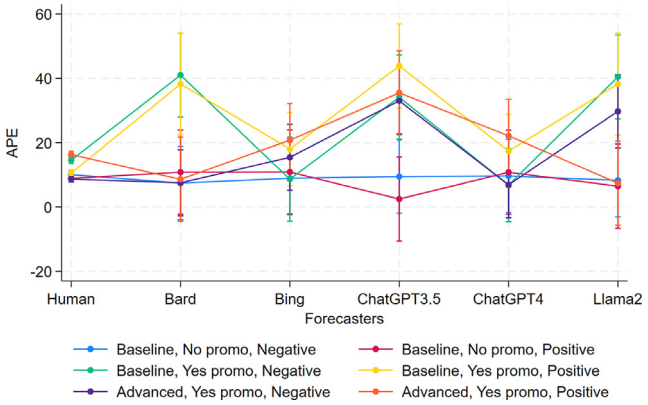


Fig. 6. Interaction: Forecasters and treatment conditions.

This suggests that advanced forecasting models, combined with positive external impacts and promotional activities, lead to decreased accuracy in human forecasts. The 'Yes promotion, Negative external impact, Baseline forecast' condition also showed an increase in APE compared to the baseline treatment condition, indicating reduced accuracy under this scenario. In contrast, the 'No promotion, Positive external impact, Baseline forecast' condition exhibited a significant decrease in APE, indicating an improvement in forecast accuracy compared to the baseline treatment condition. This suggests that human forecasters may perform better when provided with baseline forecasting under positive external impacts during non-promotional periods. This trend indicates that the interplay of promotional conditions, external impacts, and the nature of the forecasting model (baseline vs. advanced) significantly influences the accuracy of human forecasters. Human forecasters' accuracy is likely to improve under non-promotional conditions, particularly in handling positive impacts when they are provided with baseline forecasting method.

Bard showed significant decreases in APE under almost all conditions compared to the baseline treatment condition, except for the condition 'Yes promotion, Negative external impact, Baseline forecast.' Notably, the most substantial decreases were observed in 'Yes promotion, Positive external impact, Advanced forecast', 'No promotion, Positive impact, Baseline forecast', and 'No promotion, Negative impact, Baseline forecast'. This suggests that Bard performs better in terms of accuracy in these altered conditions compared to the baseline scenario. The analysis for Bing showed no significant changes in APE across all treatment conditions compared to the baseline treatment condition. The coefficients fluctuated around the baseline without showing a clear trend of increase or decrease, indicating consistent performance across different scenarios. ChatGPT3.5 demonstrated significant decreases in APE in the 'No promotion, Positive external impact, Baseline forecast' and 'No promotion, Negative impact, Baseline forecast' conditions. These results suggest improved accuracy under non-promotional conditions, irrespective of the external impact. Similar to Bing, ChatGPT4 showed no significant change in APE across different treatment conditions compared to the baseline. The coefficients indicate relatively stable performance across various scenarios. Llama2 exhibited significant decreases in APE in the 'Yes promotion, Positive impact, Advanced forecast', 'No promotion, Positive external impact, Baseline forecast', and 'No promotion, Negative impact, Baseline forecast' conditions, indicating enhanced forecasting accuracy in these scenarios compared to the baseline condition.

In our mixed-effects regression analysis investigating the difference in forecast performance by LLMs vs. human forecasters considering the complex nature of the forecasting task as represented by treatment conditions and encompassing the forecast type, promotional conditions, and external impacts (H5b), we observed significant variations in APE across different forecaster types (see Table 6 and Fig. 6). The findings suggest that, irrespective of forecaster type, the response to specific treatment conditions across all forecasters varied. In the treatment condition combining promotion with positive external impact and advanced forecasting, overall APE significantly increased. Similarly, the treatment condition with promotion, negative external impact, and baseline forecasting featured an increased APE. The non-promotional period with positive external impacts and baseline forecasting led to a decreased APE.

Regarding the main effects of individual forecaster types, significant increases in baseline APE were observed for Bard, ChatGPT3.5, and Llama2 compared to the human forecasters. No significant changes were found for Bing and ChatGPT4 vs. human forecasters.

Interaction effects between treatment conditions and forecaster types revealed notable variations, as shown in Table 6 and Fig. 6: Bard and Llama2 exhibited significant reductions in APE in several treatment conditions, particularly under advanced forecasting scenarios. Bing and ChatGPT4 displayed less variation in their performance across different treatments. Notably, forecast accuracies for ChatGPT3.5 and Llama2 were higher in non-promotional conditions under both negative and positive external impact. Generally, the findings suggest that promotional conditions typically increased the APE, with some LLMs showing a capacity to adapt more effectively to these conditions.

Table 6
Mixed-effects model results for H5.

	APE		
	Coefficient	Std. err.	P-value
Fixed effects			
Intercept	13.17	0.79	<.011
Actual sales	−0.001	0.0002	<.01
Yes promo, Positive impact, Advanced forecast	4.70	0.78	<.0011
Yes promo, Negative impact, Baseline forecast	4.54	0.75	<.01
Yes promo, Negative impact, Advanced forecast	−1.05	0.72	.142
No promo, Positive impact, Baseline forecast	−2.54	0.81	.002
No promo, Negative impact, Baseline forecast	−0.36	0.73	.620
Bard	28.48	8.13	.011
Bing	7.44	5.83	.202
ChatGPT3.5	33.22	6.68	<.01
ChatGPT4	6.82	5.83	.242
Llama2	28.48	8.13	0.01
Yes promo, Positive impact, Advanced forecast # Bard	−36.27	10.36	<.011
Yes promo, Positive impact, Advanced forecast # Bing	−2.92	8.03	.717
Yes promo, Positive impact, Advanced forecast # ChatGPT3.5	−14.14	9.27	.127
Yes promo, Positive impact, Advanced forecast # ChatGPT4	−0.93	8.03	.908
Yes promo, Positive impact, Advanced forecast # Llama2	−37.48	10.36	.011
Yes promo, Negative impact, Baseline forecast # Bard	−2.16	10.35	.835
Yes promo, Negative impact, Baseline forecast # Bing	−13.48	8.67	.120
Yes promo, Negative impact, Baseline forecast # ChatGPT3.5	−13.76	9.27	.138
Yes promo, Negative impact, Baseline forecast # ChatGPT4	−14.54	8.03	.070
Yes promo, Negative impact, Baseline forecast # Llama2	−2.73	10.35	.792
Yes promo, Negative impact, Advanced forecast # Bard	−29.64	9.49	.002
Yes promo, Negative impact, Advanced forecast # Bing	−0.67	7.62	.930
Yes promo, Negative impact, Advanced forecast # ChatGPT3.5	−8.89	8.29	.284
Yes promo, Negative impact, Advanced forecast # ChatGPT4	−8.59	7.62	.260
Yes promo, Negative impact, Advanced forecast # Llama2	−7.51	9.82	.444
No promo, Positive impact, Baseline forecast # Bard	−26.57	10.366	.010
No promo, Positive impact, Baseline forecast # Bing	−5.48	8.688	.527
No promo, Positive impact, Baseline forecast # ChatGPT3.5	−39.65	9.277	<0.01
No promo, Positive impact, Baseline forecast # ChatGPT4	−4.94	8.688	.569
No promo, Positive impact, Baseline forecast # Llama2	−30.92	10.366	.003
No promo, Negative impact, Baseline forecast # Bard	−31.13	9.833	.002
No promo, Negative impact, Baseline forecast # Bing	−8.57	8.03	.286
No promo, Negative impact, Baseline forecast # ChatGPT3.5	−33.83	8.67	<0.011
No promo, Negative impact, Baseline forecast # ChatGPT4	−7.24	8.03	.367
No promo, Negative impact, Baseline forecast # Llama2	−30.26	9.833	.002
Random effects			
	Estimate	Std. err.	
Forecaster	1.74	0.95	
Residual	127.95	3.41	
Observations	2973		

4.6. Post-experimental questionnaire: Human forecaster perspective

The post-experimental questionnaire was designed to assess the perceived importance of various information types when human participants made judgments using statistical forecasts and to evaluate their agreement with statements about the impact of promotions on their judgment. The results in Appendix K reflect a comparative assessment between baseline and advanced statistical forecasts. Participants rated the importance of information such as forthcoming promotion type, external impact, and historical sales data, with means ranging from 5.11 to 5.68, indicating a moderate to high level of importance across all factors when using both baseline and advanced forecasts. Notably, 'Advanced statistical forecast' was deemed most critical, with a mean importance rating of 5.82 when considering advanced forecasts vs. the baseline forecast (mean importance rating of 5.30). Respondents' agreement with the impact of promotions on their judgment produced mean scores around the mid-five range, indicating a mild to moderate agreement that both earlier and later promotions influenced their judgment about the forecast in Period 25. The self-assessment of the accuracy of participants' judgmental forecasts revealed a mean of 4.44 out of 7 (where 7 is 'Very high'), indicating a modestly positive self-assessment of the participants' judgmental forecasting accuracy.

5. Discussion and conclusion

This study conducted an in-depth comparison of the forecasting abilities of human experts and LLMs within the retail sector. An experimental setup was devised involving 123 human forecasters, primarily business school graduate students, and five leading LLMs—ChatGPT3.5, Bard, Llama2, Bing, and ChatGPT-4—to investigate their accuracy when predicting sales data under various conditions such as during normal and promotional sales periods. These LLMs, renowned for their advanced natural

language processing capabilities, were evaluated against human judgment, known for its understanding and adaptability in complex forecasting environments. The experiment was structured to cover diverse scenarios, encompassing both baseline and advanced forecasting models, under varying conditions like promotions and external impacts.

The results portrayed a landscape where the effectiveness of LLMs in sales forecasting was not unequivocally superior to that of human forecasters. While some LLMs, such as ChatGPT-4 and Bing, displayed forecasting accuracies on par with human participants, others like ChatGPT3.5 and Llama2 lagged behind. This outcome challenges the prevalent notion that LLMs inherently surpass human forecasters in accuracy. It stresses the importance of considering the model-specific capabilities and the context-dependent nature of these AI tools when deployed in complex forecasting scenarios. The study's findings suggest that while LLMs have considerable potential, their effectiveness is influenced by the nature of the forecasting task and the specific model's characteristics, thus requiring careful consideration when integrating them into practical forecasting processes.

5.1. Discussion of main findings

Our hypothesis that LLMs would generally surpass human forecasters in accuracy was not uniformly supported. While some LLMs, such as ChatGPT4, demonstrated higher accuracy in the forecasting experiment, its performance was not consistently better than other LLMs across various conditions. This finding is consistent with the mixed results in LLM performance reported by Schoenegger and Park (2023) and echoes Makridakis et al. (2023), who caution against assuming the uniform superiority of LLMs in forecasting. The results of this study not only reveal variations in LLM performance in forecasting but also suggest that, like humans, LLMs have differing processing capabilities that can vary from one model to another. The varied APE scores of LLMs highlight the importance of a strategic approach in selecting appropriate models for specific forecasting tasks.

The expectation that using advanced forecasting models would universally enhance human forecasting performance was not confirmed, challenging the assumption that more sophisticated models invariably improve predictions. This outcome is consistent with findings from Fildes and Goodwin (2007) and highlights the intricacies of advanced models, which may sometimes misalign with both human and AI forecasting capabilities, as discussed by Kourentzes and Petropoulos (2016). Furthermore, our findings indicate that both human and AI forecasters, including LLMs like Bard, ChatGPT3.5, and Llama2, struggle with maintaining consistent accuracy across promotional and non-promotional periods. This is particularly evident in dynamic environments where unpredictability and volatility challenge the integration of relevant information into forecasts, as noted by Lawrence et al. (2006) and Fildes and Hastings (1994). These observations suggest a need for more adaptive and intuitive forecasting systems that can better handle the complexities of such scenarios, as suggested by Sroginis et al. (2023).

The significant reduction in APE for Llama2 under positive external impact conditions partially supports our hypothesis that LLMs will exhibit varying levels of accuracy depending on the nature of external impacts—positive or negative. Although other LLMs and humans demonstrated similar accuracy under both positive and negative external impact conditions, their response patterns diverged markedly from those of Llama2. Unlike Llama2, which made minimal to no adjustments under positive conditions but significant negative adjustments under adverse conditions, human forecasters, as well as Bing and ChatGPT, responded differently. These forecasters made substantial positive adjustments in response to positive external impacts and comparatively smaller adjustments under negative conditions, as detailed in Appendix E. This varied response strategy resulted in a consistent level of accuracy across positive and negative conditions. This finding aligns with the research by Fildes et al. (2009), which suggests that larger adjustments typically lead to more significant improvements in accuracy. This is particularly evident in the case of human forecasters, whose average accuracy under these conditions was relatively better than that of ChatGPT3.5 and Llama2. Specifically, in the case of ChatGPT, which made large negative adjustments under negative external impacts and relatively small positive adjustments under positive conditions, the pattern of response underscores the complexities of forecasting accuracy across different contexts.

Looking at trends in forecast accuracy in terms of the general descriptive statistics of the median MdAPE and mean MdAPE (Appendix D), we observe that humans generally achieved the lowest median and mean MdAPEs across all categories of conditions, suggesting higher overall accuracy compared to the other forecasters. ChatGPT3.5, on the other hand, shows remarkably higher errors, particularly under promotional and advanced forecasting conditions. In contrast, Bard and Bing demonstrate substantially higher accuracy, with particularly low errors for Bard in advanced forecast conditions, suggesting its strength in more complex scenarios.

We also looked at the average forecast accuracy of various forecasters including the experiment participants, experts, and LLMs, as well as their adjustment size to provided forecasts. The results are summarized in Appendices C, D, and E. Looking at trends in forecast accuracy in terms of general descriptive statistics of the median MdAPE and mean MdAPE (Appendix D), we observe that the humans generally achieved the lowest median and mean MdAPEs across all categories of conditions, suggesting higher overall accuracy compared to the other forecasters. This is in line with the findings by Schoenegger et al. (2024), who showed that the collective wisdom of the crowd outperforms LLMs in various decision-making scenarios. Additionally, forecast errors tended to increase for almost all forecasters under promotional conditions, with ChatGPT3.5 showing the largest increase. In contrast, humans and ChatGPT4 maintained relatively lower errors, effectively managing the complexities associated with promotional effects. In the absence of promotions, most models demonstrated decreased mean and median MdAPE rates, with ChatGPT4 and Llama2 notably excelling, indicating better performance in stable market conditions.

Notably, under advanced forecasting conditions, Bard significantly improved, showing the lowest errors, which suggests that some models may have specialized capabilities that are context-dependent. However, under positive external impacts, most models—except for human forecasters—struggled with increased errors, highlighting a general sensitivity to favorable external stimuli.

Looking at the adjustment sizes to forecasts (Appendix E), distinct trends can be observed among different forecasters. Human forecasters typically exhibited the largest mean and median adjustments, indicating a robust and adaptive response to changing market conditions, particularly during promotional periods and under both positive and negative external impacts. In contrast, models like ChatGPT3.5 and Llama2 often showed reduced mean and median adjustments, particularly in complex and adverse conditions like advanced forecasting, promotional periods, and negative external impacts, suggesting a conservative approach. Our observations reveal that such smaller adjustments often correlate with increased error rates. This is particularly notable in models like Bard during advanced forecasting conditions where its minimal adjustment of −39.4 corresponded with poorer performance metrics. This trend suggests that overly conservative adjustments might not adequately address the complexities of certain forecasting scenarios, thereby potentially compromising accuracy. This perspective is supported by Fildes et al. (2009), who argued that minor adjustments generally do not enhance forecast accuracy, and it may sometimes be preferable to maintain the original forecast in the face of minor changes. Bing and ChatGPT4 displayed variability, with Bing making large adjustments under positive external impacts but decreasing adjustments under advanced forecast conditions. ChatGPT4, similarly, adapted well under promotional and positive conditions, showing a capacity for considerable adjustments akin to human forecasters. This is in line with the findings by Fildes et al. (2009), who found that large adjustments by human experts are often useful to improve the forecast accuracy. Our results show that positive adjustment is associated with lower forecast error on average. However, this may not hold for different conditions. We had a limited number of periods with no adjustment, and we observed positive adjustments more frequently than negative adjustments. This could be due to the prevalence of promotion periods in the series, which naturally required upward adjustments to the forecasts. For details, see Appendix E.

5.2. Implications

This research significantly contributes to the body of forecasting literature by empirically examining and comparing the efficacy of human forecasters and LLMs in forecasting sales under different conditions. It enhances our understanding of the complex relationship between human judgment and AI in forecasting, and challenges the assumption that LLMs are inherently superior to human forecasters. The study's findings, particularly regarding the differential performance of LLMs and the complex interaction of external factors in forecasting accuracy, provide a deeper understanding of the capabilities and limitations of both human and AI forecasters. This research highlights the importance of context and model specificity in forecasting, aligning with the theoretical frameworks suggested by Fildes and Goodwin (2007) and Lawrence et al. (2006), and offers a refined perspective on the integration of human expertise with advanced AI models in forecasting.

The study's insights are particularly valuable for practitioners in the retail industry, emphasizing the importance of a carefully considered approach when integrating LLMs into forecasting processes. The findings suggest that overreliance on LLMs without accounting for their limitations in certain contexts—particularly during promotional periods, as indicated by the increased APE values—may lead to suboptimal forecasting outcomes. Given the variability in LLM performance, especially during complex promotional periods, it is imperative for industry professionals to not overly rely on these models without due consideration of their context-specific capabilities and limitations. Our research advocates for a strategic blend of AI and human judgment, tailored to the nuances of specific forecasting scenarios. Therefore, retail practitioners should consider these insights when designing and implementing forecasting systems, ensuring they are equipped to handle the complexities of various sales contexts effectively.

The implications of this study extend to policymakers and industry regulators, who should consider these findings when establishing guidelines and frameworks for the adoption and integration of AI in business forecasting practices. The study highlights the need for policy frameworks that acknowledge the limitations and strengths of both human forecasters and LLMs. Regulatory guidelines should encourage the development of AI systems that are not only advanced in terms of data processing but also capable of being effectively integrated with human judgment. This is particularly relevant in the context of the increasing reliance on AI for decision making in various industries.

5.3. Limitations

A notable limitation of this study is the reliance on graduate students as the primary human forecasters. While these participants possess academic knowledge and some practical skills in business forecasting, they may not have the extensive practical experience and in-depth understanding typical of seasoned industry professionals. This gap potentially affects the generalizability of the findings, as real-world business forecasting often involves complex decision-making and judgment calls that are honed through extensive professional experience. The diversity in expertise and approach that experienced professionals bring to forecasting could yield different insights, especially in terms of how they interact with and interpret the outputs of advanced models like LLMs. Hence, while the study provides valuable insights into the comparative abilities of novice human forecasters and LLMs, it may not fully reflect the forecasting landscape in a professional setting.

The other limitation of this study concerns the specific data used. We provided 24 time series from real-world data on fast-moving consumer goods, each reflecting different patterns in terms of trends, promotions, and other behaviors. However, it will be useful to replicate this study with a dataset that features different patterns and comes from other industries to evaluate whether the results hold. While we focused on only two variables, 'Promotion status' and 'External impact', other variables with different settings could also be considered. For instance, in our experiment, 'External impact' was treated as a binary variable with either a 'Positive' or 'Negative' effect, capable of altering sales by up to 10%. This design was inspired by a case study; however, this variable could take different forms and have varying impacts on sales, which could be explored in future research. The study also revealed a limitation in the generalizability of its findings across various AI models due to the inconsistent performance of different LLMs. This variability necessitates a more in-depth exploration of the specific traits and training that contribute to the forecasting efficacy of these models. The LLMs selected for the study, including ChatGPT4, represent only a fraction of the rapidly evolving landscape of AI forecasting tools. Future advancements in AI technology and new model developments could significantly alter the efficacy landscape, leading to different outcomes from those observed in the current study. This limitation emphasizes the need for continuous evaluation of AI models in forecasting, considering the evolving nature of AI capabilities and the unique characteristics of each model that may impact its performance in various forecasting scenarios.

While LLMs have shown remarkable capabilities at generating human-like text and understanding context, their application in forecasting presents specific challenges—one of the most notable being the tendency for so-called hallucinations, or the generation of plausible but factually incorrect information. This phenomenon can originate from the models' reliance on the provided training data, without an inherent understanding of truth or the ability to verify facts against real-world developments (Bender, Gebru, McMillan-Major, & Shmitchell, 2021; Ribeiro, Wu, Guestrin, & Singh, 2020). Consequently, while LLMs can extrapolate trends and patterns, their predictive outputs may be compromised by inaccuracies or fabrications that may seem consistent yet are disconnected from reality. Moreover, the absence of real-time data integration further limits their forecasting reliability, as they cannot accommodate information after their training. These limitations require careful consideration and additional validation when using LLMs for forecasting purposes, highlighting the need for human oversight and the incorporation of current, domain-specific data.

5.4. Future research

The limitations identified call for future research that incorporates a broader spectrum of human forecasters, particularly those with extensive professional experience in the retail sector. Engaging experienced industry professionals would enhance the external validity of the findings and provide insights into how seasoned forecasters interact with and interpret the outputs of AI models like LLMs. Additionally, future studies should investigate a broader range of forecasting scenarios, particularly those that more closely mimic real-world complexities and uncertainties. This approach would help to understand how both human forecasters and LLMs perform under various market conditions, such as fluctuating demand, unexpected market shifts, and significant events. By examining a more diverse set of scenarios, researchers can gain a deeper understanding of the strengths and weaknesses of both human and AI forecasters in different contexts.

Further research is essential to explore how various training datasets, model architectures, and contextual inputs influence the forecasting accuracy of different LLMs, especially in complex retail environments. Investigating these aspects could reveal critical insights into the specific characteristics that enhance the forecasting capabilities of LLMs. Future studies should investigate how these two approaches can complement each other most effectively, potentially through the development of hybrid forecasting models. Such models could dynamically adjust the contribution of human and AI elements based on specific conditions, optimizing accuracy and reliability in diverse forecasting scenarios. Building on the exploratory findings from Schoenegger et al. (2024), which highlighted the potential of LLMs to significantly enhance human forecasting accuracy even when models are intentionally biased, future research should delve deeper into the mechanisms by which LLMs influence human judgment and decision-making in forecasting tasks. This research could explore the balance between AI's data-processing capabilities and human intuition and expertise, aiming to develop forecasting systems that harness the strengths of both while mitigating their limitations.

To further extend the findings of this study, future research should explore a wider array of LLMs, including emerging models with potentially different capabilities and limitations. Alongside this, the application of advanced analytical techniques could be explored, such as machine learning algorithms capable of handling nonlinear relationships and high-dimensional data. In our experiment, we provided the LLMs with relatively small datasets specific to the case study, even though these models were originally trained offline on vast amounts of data. Future studies might consider developing specialized LLMs using in-house data and tailored rules to mimic human behavior and reasoning. Such a system could serve as a decision support tool, where experts log their data—including adjustments, accuracy performance, and the rationale for those adjustments—which can be referenced in future decision-making when required. There is also potential for developing prompt-based forecasting models that can be built using companies' in-house data to generate tailored forecasts. Although this is a challenging task, the practice of using simple models already exists. Adopting this approach would deepen our understanding of how different AI models perform across various forecasting tasks, offering valuable insights into the evolution of AI in forecasting. It would also contribute to the ongoing discourse on integrating AI with human expertise in the development of advanced forecasting systems and strategies.

Finally, future studies should explore different configurations of LLM interaction—from passive to highly interactive—to assess their impact on the range and variance of predictions. This is essential for understanding whether LLMs might undermine the wisdom-of-the-crowd effect, which is critical in many forecasting contexts (Schoenegger et al., 2024). Further research should also examine the long-term effects of repeated LLM use on human forecasting skills. Investigating whether reliance on LLMs enhances or diminishes these skills over time could inform how best to integrate LLMs in practice without compromising human expertise (Schoenegger et al., 2024).

5.5. Concluding remarks

This study's examination of the forecasting abilities of both human experts and LLMs in retail sales yielded several key insights. The experiments, which involved human forecasters—primarily business school graduate students—and five leading LLMs, including ChatGPT4, demonstrated that LLM performance in forecasting is not consistently superior to that of humans. While some models, like ChatGPT4 and Bing, showed forecasting accuracy comparable to human forecasters, others, such as ChatGPT3.5 and Llama2, did not perform as well. These findings indicate that the effectiveness of LLMs in forecasting tasks depends on the specific model and the nature of the task.

The study highlighted that advanced forecasting models do not automatically improve forecasting performance. Both human forecasters and LLMs exhibited limitations when using these models, suggesting that integrating sophisticated tools into forecasting requires careful consideration. This finding aligns with previous research indicating that the complexity of advanced models can sometimes outweigh their potential benefits in accuracy. Additionally, the research revealed that both human and AI forecasters struggle with the unpredictability of promotional periods, as evidenced by increased APE values during these times. This highlights a shared challenge in adapting to dynamic market conditions and emphasizes the importance of further research to explore how human judgment and advanced AI models can be integrated to optimize their combined strengths in forecasting

CRediT authorship contribution statement

Mahdi Abolghasemi: Conceptualization, Methodology, Software, Formal analysis, Validation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Odkhishig Ganbold:** Conceptualization, Methodology, Software, Formal analysis, Validation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Kristian Rotaru:** Conceptualization, Methodology, Validation, Data curation, Writing – original draft, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

Data and code are available on the following GitHub repository https://github.com/mahdiabolghasemi/LLM-vs.-Humans-Forecasting.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2024.07.003.

References

Abolghasemi, M., Beh, E., Tarr, G., & Gerlach, R. (2020). Demand forecasting in supply chain: The impact of demand volatility in the presence of promotion. *Computers & Industrial Engineering*, *142*, Article 106380.

Abolghasemi, M., Hurley, J., Eshragh, A., & Fahimnia, B. (2020). Demand forecasting in the presence of systematic events: Cases in capturing sales promotions. *International Journal of Production Economics*, Article 107892.

Abolghasemi, M., Tarr, G., & Bergmeir, C. (2022). Machine learning applications in hierarchical time series forecasting: Investigating the impact of promotions. *International Journal of Forecasting*, *40*(2), 597–615.

Arvan, M., Fahimnia, B., Reisi, M., & Siemsen, E. (2019). Integrating human judgement into quantitative forecasting methods: A review. *Omega*, *86*, 237–252.

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency* (pp. 610–623).

Bolger, F., & Harvey, N. (1993). Context-sensitive heuristics in statistical reasoning. *The Quarterly Journal of Experimental Psychology Section A*, *46*(4), 779–811.

Bolger, F., & Önkal-Atay, D. (2004). The effects of feedback on judgmental interval predictions. *International Journal of Forecasting*, *20*(1), 29–39.

Brau, R., Aloysius, J., & Siemsen, E. (2023). Demand planning for the digital supply chain: How to integrate human judgment and predictive analytics. *Journal of Operations Management*, *69*(6), 965–982.

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., ... Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems*, *33*, 1877–1901.

Chang, Y. Y., & Hao, W. (2022). Negativity bias of analyst forecasts. *Journal of Behavioral Finance*, *23*(2), 175–188.

Chang, Y., Wang, X., Wang, J., Wu, Y., Zhu, K., Chen, H., ... Xie, X. (2023). A survey on evaluation of large language models. arXiv preprint arXiv:2307.03109.

Chapman, G. B., & Johnson, E. J. (1999). Anchoring, activation, and the construction of values. *Organizational Behavior and Human Decision Processes*, *79*(2), 115–153.

Cnaan, A., Laird, N. M., & Slasor, P. (1997). Using the general linear mixed model to analyse unbalanced repeated measures and longitudinal data. *Statistics in Medicine*, *16*(20), 2349–2380.

De Baets, S., & Harvey, N. (2018). Forecasting from time series subject to sporadic perturbations: Effectiveness of different types of forecasting support. *International Journal of Forecasting*, *34*(2), 163–180.

Eroglu, C., & Croxton, K. L. (2010). Biases in judgmental adjustments of statistical forecasts: The role of individual differences. *International Journal of Forecasting*, *26*(1), 116–133.

Fildes, R. (1991). Efficient use of information in the formation of subjective industry forecasts. *Journal of Forecasting*, *10*(6), 597–617.

Fildes, R., & Goodwin, P. (2007). Against your better judgment? How organizations can improve their use of management judgment in forecasting. *Interfaces*, *37*(6), 570–576.

Fildes, R., Goodwin, P., Lawrence, M., & Nikolopoulos, K. (2009). Effective forecasting and judgmental adjustments: an empirical evaluation and strategies for improvement in supply-chain planning. *International Journal of Forecasting*, *25*(1), 3–23.

Fildes, R., Goodwin, P., & Önkal, D. (2019). Use and misuse of information in supply chain forecasting of promotion effects. *International Journal of Forecasting*, *35*(1), 144–156.

Fildes, R., & Hastings, R. (1994). The organization and improvement of market forecasting. *Journal of the Operational Research Society*, *45*(1), 1–16.

Fildes, R., Ma, S., & Kolassa, S. (2022). Retail forecasting: Research and practice. *International Journal of Forecasting*, *38*(4), 1283–1318.

Goodwin, P. (2005). Providing support for decisions based on time series information under conditions of asymmetric loss. *163*, (2), (pp. 388–402).

Goodwin, P., & Fildes, R. (2022). Forecasting with judgment. In *The palgrave handbook of operations research* (pp. 541–572). Cham: Springer International Publishing.

Goodwin, P., & Wright, G. (2010). The limits of forecasting methods in anticipating rare events. *Technological Forecasting and Social Change*, *77*(3), 355–368.

Green, K. C., & Armstrong, J. S. (2015). Simple versus complex forecasting: The evidence. *Journal of Business Research*, *68*(8), 1678–1685.

Gueorguieva, R., & Krystal, J. H. (2004). Move over ANOVA: Progress in analyzing repeated-measures data and its reflection in papers published in the archives of general psychiatry. *Archives of General Psychiatry*, *61*(3), 310–317.

Halawi, D., Zhang, F., Yueh-Han, C., & Steinhardt, J. (2024). Approaching human-level forecasting with language models. arXiv preprint arXiv:2402.18563.

Harvey, N. (1995). Why are judgments less consistent in less predictable task situations? *Organizational Behavior and Human Decision Processes*, *63*(3), 247–263.

Harvey, N., & Bolger, F. (1996). Graphs versus tables: Effects of data presentation format on judgemental forecasting. *International Journal of Forecasting*, *12*(1), 119–137.

Ito, T. A., Larsen, J. T., Smith, N. K., & Cacioppo, J. T. (1998). Negative information weighs more heavily on the brain: The negativity bias in evaluative categorizations. *Journal of Personality and Social Psychology*, *75*(4), 887.

Kim, A., Muhn, M., & Nikolaev, V. (2024). Financial statement analysis with large language models. arXiv preprint arXiv:2407.17866.

Kirshner, S. (2024). Artificial agents and operations management decision-making. Available at SSRN: http://dx.doi.org/10.2139/ssrn.4726933.

Kourentzes, N., & Fildes, R. (2023). The dynamics of judgemental adjustments in demand planning. Available at SSRN 4534744. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4534744.

Kourentzes, N., & Petropoulos, F. (2016). Forecasting with multivariate temporal aggregation: The case of promotional modelling. *International Journal of Production Economics*, *181*, 145–153.

Lawrence, M., Goodwin, P., O'Connor, M., & Önkal, D. (2006). Judgmental forecasting: A review of progress over the last 25 years. *International Journal of Forecasting*, *22*(3), 493–518.

Lewandowsky, S., Ecker, U. K., Seifert, C. M., Schwarz, N., & Cook, J. (2012). Misinformation and its correction: Continued influence and successful debiasing. *Psychological Science in the Public Interest*, *13*(3), 106–131.

Li, X., Feng, H., Yang, H., & Huang, J. (2024). Can chatgpt reduce human financial analysts' optimistic biases? *Economic and Political Studies*, *12*(1), 20–33.

Lim, J. S., & O'Connor, M. (1995). Judgemental adjustment of initial forecasts: Its effectiveness and biases. *Journal of Behavioral Decision Making*, *8*(3), 149–168.

Ma, S., Fildes, R., & Huang, T. (2016). Demand forecasting with high dimensional data: The case of SKU retail sales forecasting with intra- and inter-category promotional information. *European Journal of Operational Research*, *249*, 245–257.

Makridakis, S., Petropoulos, F., & Kang, Y. (2023). Large language models: Their success and impact. *Forecasting*, *5*(3), 536–549.

Meub, L., & Proeger, T. (2016). Can anchoring explain biased forecasts? Experimental evidence. *Journal of Behavioral and Experimental Finance*, *12*, 1–13.

Önkal, D., Gönül, M. S., & Goodwin, P. (2023). Supporting judgment in predictive analytics: Scenarios and judgmental forecasts. In *Judgment in predictive analytics* (pp. 245–264). Cham: Springer International Publishing.

Perera, H. N., Hurley, J., Fahimnia, B., & Reisi, M. (2019). The human factor in supply chain forecasting: A systematic review. *European Journal of Operational Research*, *274*(2), 574–600.

Reimers, S., & Harvey, N. (2023). Bars, lines and points: The effect of graph format on judgmental forecasting. *International Journal of Forecasting*, *40*(1), 44–61.

Ribeiro, M. T., Wu, T., Guestrin, C., & Singh, S. (2020). Beyond accuracy: Behavioral testing of NLP models with CheckList. arXiv preprint arXiv:2005.04118.

Schoenegger, P., & Park, P. S. (2023). Large language model prediction capabilities: Evidence from a real-world forecasting tournament. arXiv preprint arXiv:2310.13014.

Schoenegger, P., Park, P. S., Karger, E., & Tetlock, P. E. (2024). AI-augmented predictions: LLM assistants improve human forecasting accuracy. arXiv preprint arXiv:2402.07862.

Schoenegger, P., Tuminauskaite, I., Park, P. S., & Tetlock, P. E. (2024). Wisdom of the silicon crowd: Ensemble prediction capabilities rival human crowd accuracy. arXiv preprint arXiv:2402.19379.

Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., Van Den Driessche, G., ... Hassabis, D. (2016). Mastering the game of go with deep neural networks and tree search. *Nature*, *529*(7587), 484–489.

Sroginis, A., Fildes, R., & Kourentzes, N. (2023). Use of contextual and model-based information in adjusting promotional forecasts. *European Journal of Operational Research*, *307*(3), 1177–1191.

Trapero, J. R., Pedregal, D. J., Fildes, R., & Kourentzes, N. (2013). Analysis of judgmental adjustments in the presence of promotions. *International Journal of Forecasting*, *29*(2), 234–243.

Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases: Biases in judgments reveal some heuristics of thinking under uncertainty. *Science*, *185*(4157), 1124–1131.

Webby, R., O'Connor, M., & Edmundson, B. (2005). Forecasting support systems for the incorporation of event information: An empirical investigation. *International Journal of Forecasting*, *21*(3), 411–423.

Xue, H., Voutharoja, B. P., & Salim, F. D. (2022). Leveraging language foundation models for human mobility forecasting. In *Proceedings of the 30th international conference on advances in geographic information systems* (pp. 1–9).

Yu, X., Chen, Z., Ling, Y., Dong, S., Liu, Z., & Lu, Y. (2023). Temporal data meets LLM—explainable financial time series forecasting. arXiv preprint arXiv:2306.11025.

Zhang, G., Patuwo, B. E., & Hu, M. Y. (1998). Forecasting with artificial neural networks: The state of the art. *International Journal of Forecasting*, *14*(1), 35–62.

Zhao, H., Liu, Z., Wu, Z., Li, Y., Yang, T., Shu, P., ... Liu, T. (2024). Revolutionizing finance with llms: An overview of applications and insights. arXiv preprint arXiv:2401.11641.